

Questions by topic: How humans influence biodiversity - Set 1 (45 marks)

Answer all questions in the spaces provided.

01 . 1

For each of the following forms of direct exploitation, give a named species whose population has been reduced.

[5 marks]

Food

.....

Fashion

.....

Entertainment / pets

.....

Furniture and ornaments

.....

Traditional medicines

.....

01 . 2

Explain why species are deliberately eradicated. Give **three** categories with examples.

[3 marks]

1

.....

2

.....

3

.....

01 . 3

Explain why legal protection alone rarely prevents the decline of a species exploited for traditional medicine.

[2 marks]

.....

.....

10

0 2 . 1

Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

[6 marks]

Water availability

.....

.....

.....

Dissolved oxygen

.....

.....

.....

pH

.....

.....

.....

0 2 . 2

Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

[2 marks]

.....

.....

.....

0 2 . 3

Explain why species that were already rare are most likely to be lost when abiotic conditions change.

[2 marks]

.....

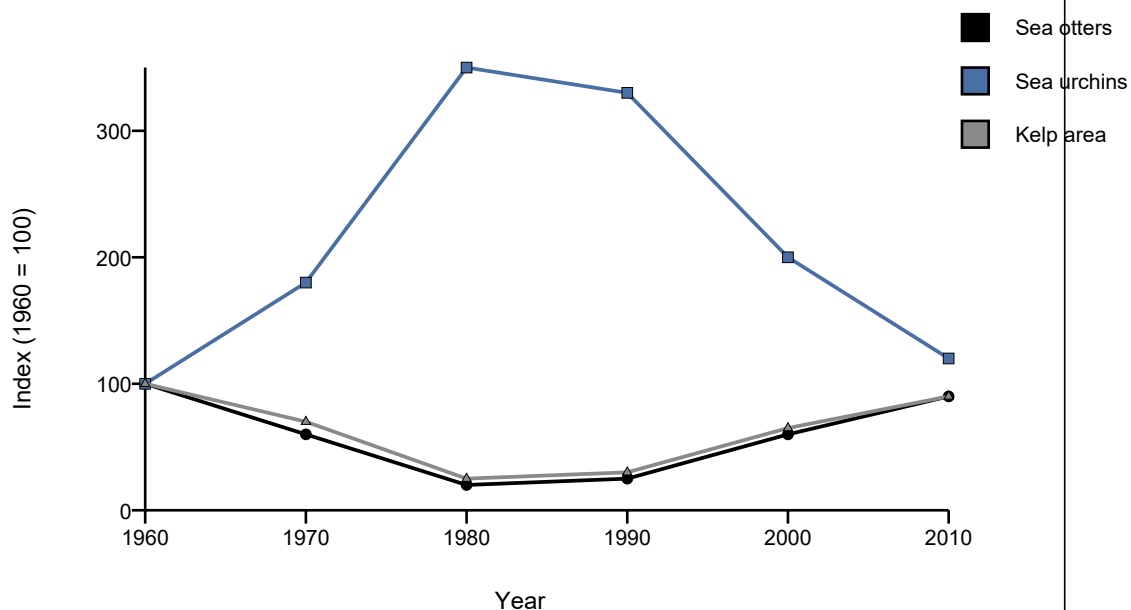
.....

10

03

Figure 1 shows the estimated populations of sea otters, sea urchins and the area of kelp forest along part of the coast of California between 1960 and 2010.

Figure 1



03

. 1

Describe the relationship between the three variables shown in Figure 1.

[3 marks]

03

. 2

Explain the relationships shown in Figure 1.

[3 marks]

03 . 3

Give **two** other examples of human over-exploitation of one species affecting the population of another.

[2 marks]

1

2

03 . 4

Explain why the loss of beavers or forest elephants can cause the decline of many other species.

[2 marks]

10

04 . 1

Explain why introduced species often have a greater effect on island communities than on mainland communities.

[2 marks]

04 . 2

For each of the following, describe an example of an introduced species and explain its effect on an indigenous species.

[6 marks]

Introduced competitor

.....

.....

.....

Introduced predator

.....

.....

.....

Introduced pathogen

.....

.....

.....

0 4 . 3

Explain how hybridisation with an introduced species threatens the red deer and the Scottish wildcat.

[2 marks]

.....

.....

.....

.....

10

0 5 . 1

Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

- A Thermal pollution of a river by a power station
- B Flooding a valley to create a reservoir
- C Lowering the water table by groundwater abstraction
- D Acid mine drainage entering a stream

☐

☐

☐

☐

0 5 . 2

Give **four** human activities that cause complete habitat destruction.

[4 marks]

1

.....

2

.....

3

.....

4

.....

5

END OF QUESTIONS